The Performance of a Type of Swirl Atomizer*

By A. Radcliffe, B.Sc.†

INTRODUCTION

Fuel is frequently injected into gas turbines through swirl jets. The principles of operation of all swirl jets are the same, but the variation of performance from one design to another is such that it is usually difficult for those not actively engaged in the manufacture of a particular type of jet to estimate the performance.

Fig. 1*a* shows one arrangement of an atomizer. It consists of a main body and a plug. There are flats on the side of the atomizer so arranged that when the body is screwed into a fuel delivery line, the fuel has unrestricted access to each of the tangential ports. Fig. 1*b* shows another arrangement of the atomizer. In this instance the orifice plate and back plug make a push fit with the swirl sleeve. The three pieces when assembled fit into a body so arranged that fuel has access to the two swirl ports. The essential details of the atomizer are shown in Fig. 1*c*. In brief, the arrangement is such that fuel enters the swirl chamber through two ports of diameter, p. Each port admits fuel into the swirl chamber which may be regarded as a very short cylinder of diameter, D, and length, b, leading to a cone of included angle, ϕ. The fuel flows in a vortex in the chamber leaving through the final orifice of diameter, d. The flow mechanism has been explained (Taylor 1950)‡. When the atomizer is operating satisfactorily there is an air core at the centre of the vortex so that fuel passes through the orifice as a thin film with axial and tangential velocities. On leaving the orifice the fuel spreads as a cone of included angle, θ. The surface of the cone tends to form transverse waves and break up into filaments and droplets (Squire 1953). Should the fuel pressure fall, or the fuel viscosity rise, too far, the surface tension forces acting in the cone may pull it into the shape of a tulip. In worse conditions the air core disappears and the fuel issues from the orifice as one or more filaments of fuel. The atomization in such instances is poor.

Simons (1946) in unpublished work found that the atomizer operated satisfactorily when the relation between the various dimensions was as set out in Table 1.

TABLE 1. DETAILS OF ATOMIZER OF STANDARD SHAPE

Diameter of final orifice	d
Diameter of tangential ports	$p = d$
Length of final orifice	$t = 2d/3$
Diameter of swirl chamber	$D = 6d$ to $9d$
Length of cylindrical part of swirl chamber	$b = 2d$
Radius of offset of tangential ports	$r = D/2-3d/4$
Angle of cone of swirl chamber	$\phi = 60$ to 90 deg.

The dimensions as quoted are mostly referred to the final orifice diameter, d. The latitude in the choice of D and ϕ is useful in practical cases. For the purposes of this paper a standard shape of atomizer is defined as one with the dimensions as tabulated above except that D is fixed at $7{\cdot}5d$, and ϕ at 90 deg.

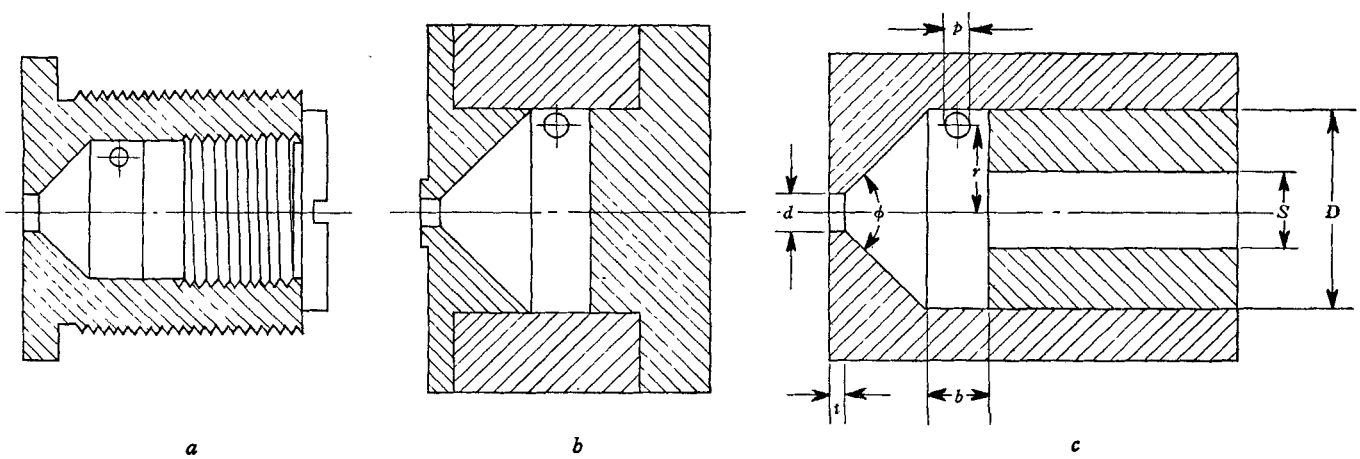

Fig. 1. Arrangements of Atomizer

- b Length of cylindrical part of swirl chamber.
- D Diameter of swirl chamber.
- d Diameter of final orifice.
- p Diameter of tangential ports.
- r Radius of offset of tangential ports.
- S Diameter of spill orifice.
- t Length of final orifice.
- ϕ Angle of cone of swirl chamber.

FUEL FLOW FOR ATOMIZERS OF STANDARD SHAPE

The fuel flow through atomizers of a given shape has long been known to be dependent upon the atomizer size, the applied pressure and the viscosity and density of the fuel. A dimensional analysis of the problem shows that, if the various parameters are represented by symbols as in Table 2, then

$$\frac{Q^2}{d^4P\rho} = f\left(\frac{Q}{\mu d}\right) \quad . \quad . \quad . \quad . \quad . \quad . \quad (1)$$

Preliminary tests with liquids of similar densities but different viscosities (that is, various mixtures of kerosene and hydraulic oil, D.T.D. 44D), and with fuels of different density but similar viscosity (that is, mixtures of kerosene and carbon tetrachloride) were made on one particular atomizer and it was shown that an equation such as (1) represented the flow through an atomizer. A series of atomizers of similar shape but different size were then

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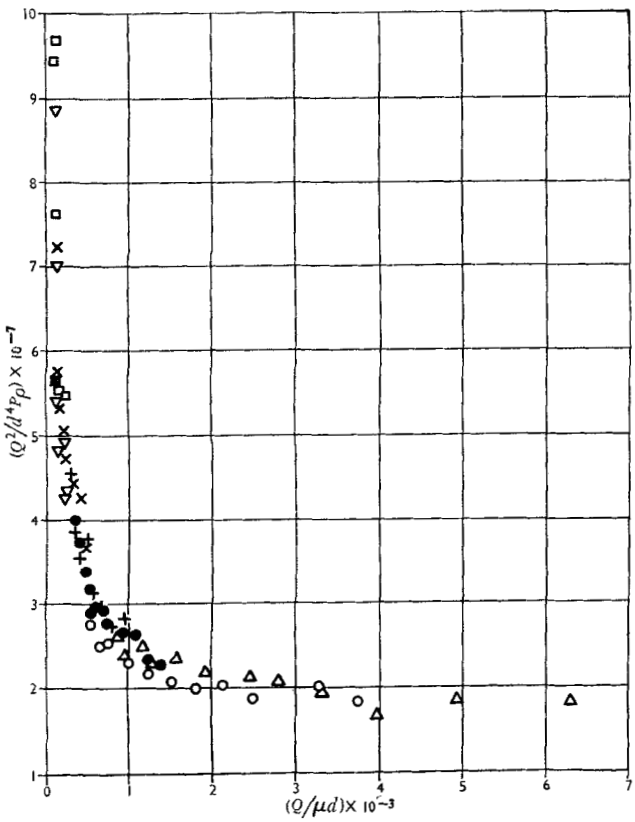
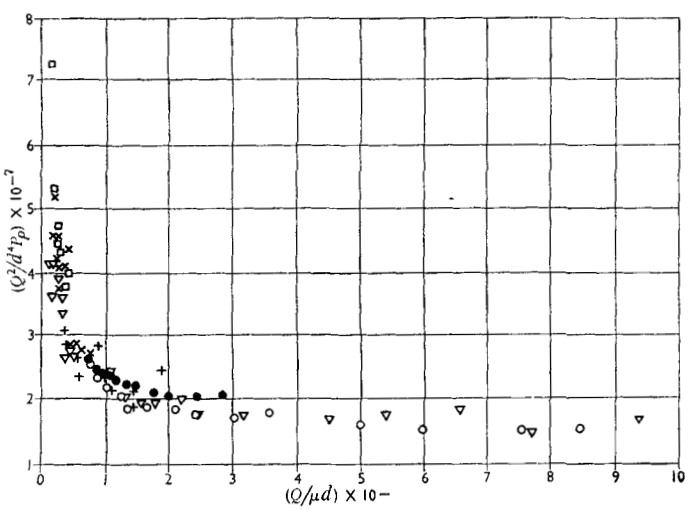
† National Gas Turbine Establishment, Ministry of Supply.

‡ An alphabetical list of references is given in the Appendix.

manufactured and the flow through these atomizers was measured for seven different fluids. The fluids were (1) carbon tetrachloride, (2) petrol, (3) kerosene, and (4)–(7) four mixtures of kerosene and hydraulic oil. The density range covered was 0·75–1·6 grammes per cu. cm., the viscosity range was 0·5–25 centipoises and the pressure range was 5–1,000 lb. per sq. in.



a For atomizer (3).



b For atomizer (6).

Fig. 2. $Q^2/d^4P\rho$ Plotted Against $Q/\mu d$ for Several Liquids

- ● Kerosene.
- △ Petrol.
- ○ Carbon tetrachloride.
- + One part of D.T.D. 44D to one part of kerosene.
- × Three parts of D.T.D. 44D to one part of kerosene.
- ▽ Seven parts of D.T.D. 44D to one part of kerosene.
- □ Fifteen parts of D.T.D. 44D to one part of kerosene.

TABLE 2. PARAMETERS CONTROLLING FLOW IN SWIRL ATOMIZERS

Parameter	Symbol	Dimension
Flow rate	Q	M/T
Pressure	P	M/LT^2
Diameter of orifice	d	L
Density of fuel	ρ	M/L^3
Viscosity	μ	M/LT

The atomizers had final orifices of nominal diameter 0·020, 0·030, 0·040, 0·050, 0·060, and 0·070 inch and were numbered (2)–(7) respectively. The measured dimensions of these atomizers are listed in Table 3.

TABLE 3. DIMENSIONS OF ATOMIZERS OF SIMILAR SHAPE

(Measurements in inches.)

Atomizer No.	Orifice diameter, d	Orifice length, t	Port diameter, p	Chamber diameter, D	Chamber length, b	Offset radius, r
(2)	0·0184	0·010	0·0226	0·150	0·040	0·058
(3)	0·0285	0·015	0·0304	0·219	0·058	0·087
(4)	0·0406	0·020	0·0419	0·299	0·090	0·118
(5)	0·0514	0·025	0·0527	0·389	0·103	0·155
(6)	0·0588	0·030	0·0616	0·443	0·117	0·175
(7)	0·0692	0·035	0·0732	0·520	0·141	0·205

For each test the volume liquid-flow, liquid temperature, and applied fuel pressure were measured. The temperature measurement enabled the liquid viscosity and density to be calculated from previously measured variations of these quantities with temperature. Thus for each test it was possible to determine Q in lb. per hr., P in lb. per sq. in., d in inches, ρ in grammes per cu. cm., and the viscosity, μ, in centipoises. The units used are, of course, inconsistent but they are those likely to be used in Britain by users of fuel atomizers. The values of $Q^2/d^4P\rho$ and $Q/\mu d$ were calculated and plotted for each of the atomizers (2)–(7). The points for each atomizer lay close to a common curve. Fig. 2*a* and *b* shows the data for atomizers (3)

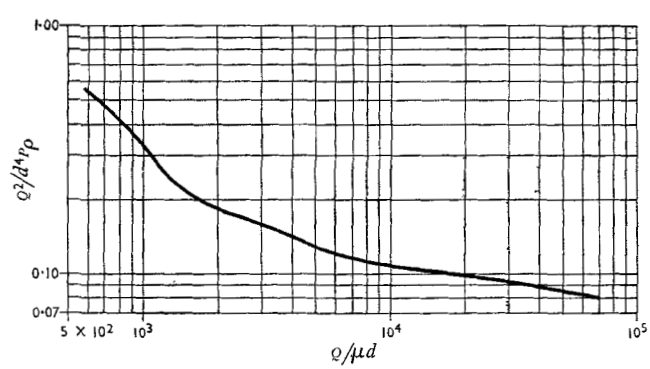


Fig. 3. Relation Between $Q^2/d^4P\rho$ and $Q/\mu d$ (Consistent Units)

and (6). Fig. 3 shows the best curve through all the points for the six atomizers. The values of $Q^2/d^4P\rho$ and $Q/\mu d$ have been reduced to those for consistent units. The errors in measuring Q, P, ρ, and μ, and in assuming that d represents the size of each atomizer, varied with the test conditions, but the standard deviation was of the order of 1 per cent for all the measurements except ρ. Random errors of this order readily explain the scatter of the points.

By using the curve of Fig. 3, the relation between $Q^2/P\rho$ and Q/μ for any value of d may be deduced. Fig. 4 shows ten such curves plotted on a logarithmic scale for values of d in the range 0·010–0·100 inch. By use of these curves it is possible by

interpolation to determine the orifice diameter and, hence, all other dimensions of the standard atomizer needed to satisfy any particular set of a wide range of flow, pressure, density, and viscosity conditions. The scope of the curves can be extended to serve for higher values of $Q^2/P\rho$ and Q/μ by dividing the former by $10{,}000^m$, the latter by 10^m, and multiplying the value of d so discovered by 10^m, where m is positive. It will be noted that the

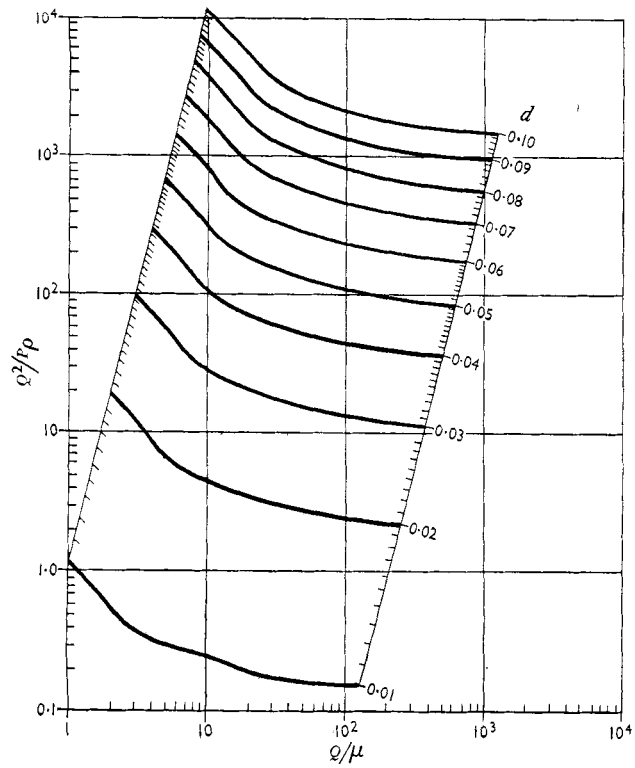


Fig. 4. Relation Between $Q^2/P\rho$ and Q/μ for Various Values of d

Q is measured in lb. per hr.; P in lb. per sq. in.; d in inches; μ in centipoises; and ρ in grammes per cu. cm.

graphs of Figs. 3 and 4 have a marked change of slope for values of $Q/\mu d$ in the region of 1,600 measured in consistent units. As the value of $Q/\mu d$ is reduced the atomizer tends to fail to operate properly because the bubble does not open or is not properly formed. $Q/\mu d$ can only be reduced to give the part of the graph to the left of the change of slope if this is done by increasing μ to as much as 30 centipoises. If both $Q/\mu d$ and μ are low the atomizer does not operate with a conical spray. It is, for example, unlikely that an atomizer will operate for a value of $Q/\mu d$ less than 1,600 when μ is as small as 1 centipoise. There is no apparent reason why the curves of Fig. 4 should not be extended to higher values of Q/μ.

It is interesting to note that Fig. 4 shows that the flow number (usually defined as the flow in gallons per hour divided by the square root of pressure in lb. per sq. in.) is not constant. Indeed, it varies with density, viscosity, and fuel pressure. However, as it has its uses, the relation between the flow number and orifice diameter of the standard atomizer, where the fuel is kerosene of density 0·800 gramme per cu. cm. and viscosity 2·15 centipoises, is shown in Fig. 7. The pressure at which the measurements of fuel flow were made was 100 lb. per sq. in.

THE EFFECT OF ATOMIZER SHAPE ON PERFORMANCE

To determine how atomizer shape affected the flow rate and spray-cone angle a set of atomizers was made in accordance with the arrangements shown in Fig. 1*b*. Thirty-one swirl sleeves, nine orifice plates, one back stop, and one body were manufactured. The swirl sleeves and orifice plates were interchangeable so that a total of 279 different atomizers could be assembled. The

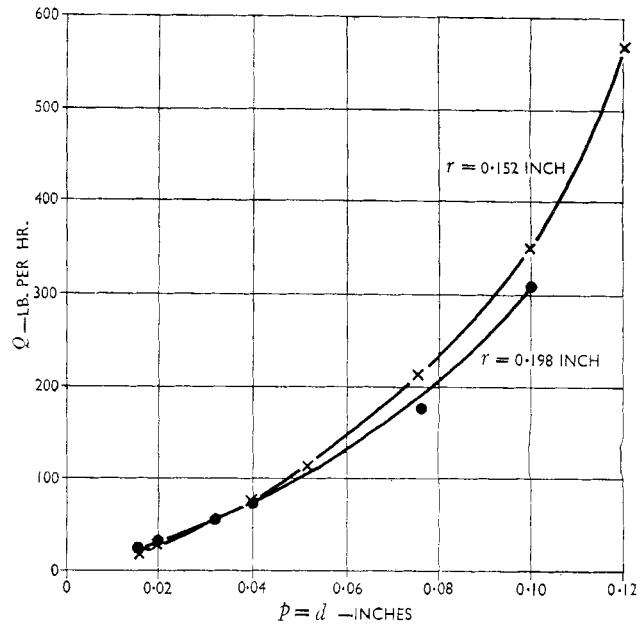


Fig. 5. Variation of Fuel Flow as Port and Final Diameter are Both Changed

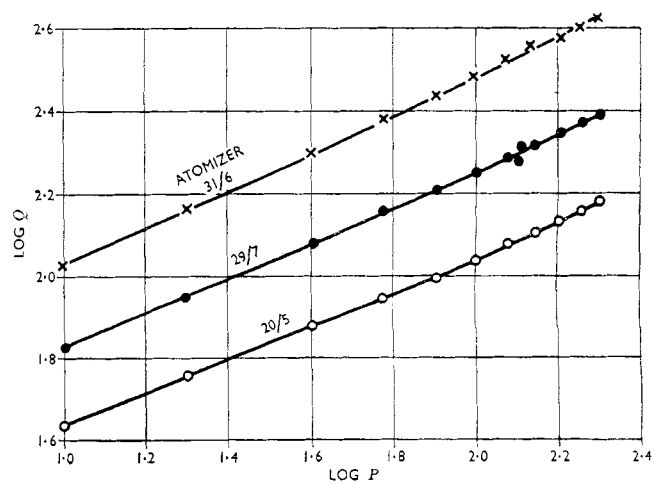


Fig. 6. Variation of Kerosene Flow with Pressure for Three Dissimilar Atomizers

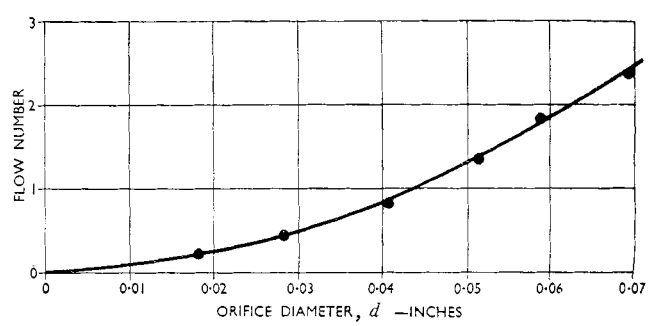


Fig. 7. Variation of Flow Number with Orifice Diameter for Kerosene

Flow number is equal to one-tenth of the fuel flow in gal. per hr. at a pressure of 100 lb. per sq. in.

orifice plates each had a 90 deg. cone of maximum diameter, D, of 0·500 inch, but had different orifices of diameter and length as listed in Table 4. The thirty-one swirl sleeves had an internal diameter of 0·500 inch. The port diameter, p, and the port offset radius, r, were as listed in Table 5. The swirl sleeves, plates, and back stop were so made that the length of the cylindrical part of the swirl chamber was 0·1125 inch. The swirl sleeves were numbered from (1) to (31) and the orifice plates from (1) to (9) and the various combinations were indicated by writing the

TABLE 4. DIMENSIONS OF ORIFICE PLATES

Orifice plate No.	Orifice diameter, d, inch	Orifice length, t, inch
(1)	0·016	0·015
(2)	0·0195	0·020
(3)	0·033	0·0305
(4)	0·042	0·039
(5)	0·052	0·053
(6)	0·067	0·040
(7)	0·076	0·044
(8)	0·100	0·063
(9)	0·120	0·065

sleeve number first and the plate number last (for example, (27/8)). Each arrangement was tested for fuel flow and cone angle. The fuel was kerosene of density 0·800 and viscosity 2·15 centistokes. The fuel flow was measured in rotameters calibrated by displacement methods. The cone angle was determined by projecting a silhouette of the spray on to a ground-glass screen: this method really measures the cone angle of the envelope of

TABLE 5. DIMENSIONS OF SWIRL SLEEVES

Sleeve No.	Port diameter, p, inch	Offset radius, r, inch	Sleeve No.	Port diameter, p, inch	Offset radius, r, inch
(1)	0·016	0·050	(17)	0·020	0·198
(2)	0·020	0·049	(18)	0·031	0·198
(3)	0·031	0·0495	(19)	0·040	0·1985
(4)	0·040	0·0495	(20)	0·052	0·1985
(5)	0·052	0·0495	(21)	0·077	0·095
(6)	0·016	0·096	(22)	0·083	0·0935
(7)	0·020	0·096	(23)	0·105	0·095
(8)	0·031	0·0965	(24)	0·122	0·096
(9)	0·040	0·097	(25)	0·077	0·1505
(10)	0·052	0·097	(26)	0·084	0·151
(11)	0·016	0·152	(27)	0·102	0·148
(12)	0·020	0·152	(28)	0·122	0·150
(13)	0·031	0·1515	(29)	0·076	0·1985
(14)	0·040	0·152	(30)	0·083	0·198
(15)	0·052	0·151	(31)	0·100	0·197
(16)	0·016	0·198			

the spray. The pressure for fuel-flow and cone-angle tests was, in each instance, 100 lb. per sq. in. Additionally, the atomizers (20/5), (29/7), and (31/8), which most closely approached the design suggested by Simons, were tested for fuel flow at pressures of 10–200 lb. per sq. in. Table 6 lists the cone angle and flow rate through the 279 atomizers.

ATOMIZER SHAPE AND ITS EFFECT OF FUEL FLOW

Figs. 5, 8, and 9 show some of the many curves which may be drawn to represent the flow results graphically. Fig. 8 shows how the fuel flow varies with orifice diameter when the port diameter is held constant. In the range where $d = D/7{\cdot}5$ the fuel flow is proportional to $d^{0{\cdot}94}$. Fig. 9 similarly shows how the fuel flow varies with port diameter. The relation is $Q \propto p^{0{\cdot}70}$. If the orifice and port diameters are varied together it would be expected that the fuel flow, Q, would be proportional to d or p raised to the power 1·64. Fig. 5 shows results which agree with this within the limits of experimental error. For the standard atomizer it is possible to show from Fig. 3 that when the applied fuel pressure is 100 lb. per sq. in. and d, p, and $D/7{\cdot}5$ are each of the order of 0·067 inch, then $Q \propto d^{1{\cdot}93}$. Thus the effect of varying linear dimensions, l, which are not varied automatically when the orifice and port diameters are changed, is such as to cause a fuel-

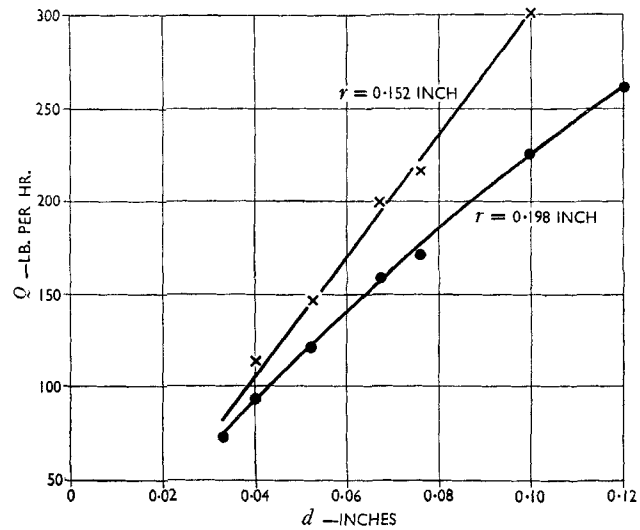


Fig. 8. Variation of Fuel Flow with Final Orifice Diameter

flow variation proportional to $l^{0{\cdot}29}$. The effect of altering atomizer shape may thus be summarized in the equation

$$Q = Ad^{0{\cdot}94}p^{0{\cdot}70}l^{0{\cdot}29} \quad . \quad . \quad . \quad . \quad (2)$$

where A is a constant. This relation only applies accurately over a limited range but will always give an approximate method of finding what happens when burner shape is changed.

Fig. 6 shows the variation of fuel flow with pressure for three dissimilar atomizers. It is notable that the ratio of the fuel flows changes very little with pressure.

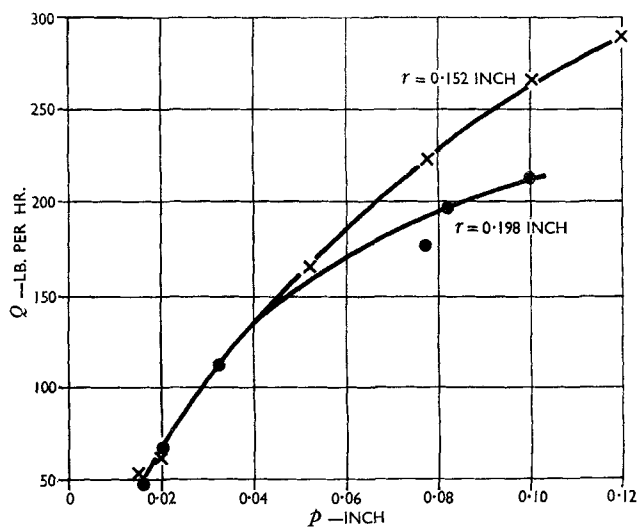


Fig. 9. Variation of Fuel Flow with Tangential Port Diameter

The fuel flow from an atomizer can be altered by varying the pressure but practical difficulties limit the possible pressure ratio to little more than 100 to 1 and the fuel flow to a ratio not much in excess of 10 to 1. An alternative method of varying the fuel flow depends on the use of a spill orifice in the rear wall of the swirl chamber. Fig. 1c shows a typical arrangement where the spill orifice has a diameter, S. By altering the pressure in the

TABLE 6. FUEL FLOW AND CONE ANGLE OF 279 ATOMIZERS

Burner No.	Orifice diameter, d, inch	Port diameter, p, inch	Offset radius, r, inch	Cone angle, θ, deg.	Tan $(\theta/2)$	Flow, lb. per hr.
(1/1)	0·016	0·016	0·050	F*	—	—
(2/1)	,,	0·020	0·049	F	—	—
(3/1)	,,	0·031	0·0495	F	—	—
(4/1)	,,	0·040	0·0495	F	—	—
(5/1)	,,	0·052	0·0495	F	—	—
(6/1)	,,	0·016	0·096	F	—	—
(7/1)	,,	0·020	0·096	F	—	—
(8/1)	,,	0·031	0·0965	F	—	—
(9/1)	,,	0·040	0·097	F	—	—
(10/1)	,,	0·052	0·052	F	—	—
(21/1)	,,	0·077	0·095	F	—	—
(22/1)	,,	0·083	0·093	F	—	—
(23/1)	,,	0·105	0·095	F	—	—
(24/1)	,,	0·122	0·096	F	—	—
(11/1)	,,	0·016	0·152	53	0·499	23·5
(12/1)	,,	0·020	0·152	F	—	—
(13/1)	,,	0·031	0·1515	F	—	—
(14/1)	,,	0·040	0·162	F	—	—
(15/1)	,,	0·052	0·151	F	—	—
(25/1)	,,	0·077	0·1505	F	—	—
(26/1)	,,	0·084	0·151	F	—	—
(27/1)	,,	0·102	0·148	F	—	—
(28/1)	,,	0·122	0·150	F	—	—
(16/1)	,,	0·016	0·198	F	—	—
(17/1)	,,	0·020	0·198	F	—	—
(18/1)	,,	0·031	0·198	F	—	—
(19/1)	,,	0·040	0·1985	F	—	—
(20/1)	,,	0·052	0·1985	F	—	—
(29/1)	,,	0·076	0·1985	F	—	—
(30/1)	,,	0·083	0·198	F	—	—
(31/1)	,,	0·100	0·197	F	—	—
(1/2)	0·0195	0·016	0·050	64	0·625	28·4
(2/2)	,,	0·020	0·049	63	0·613	34·9
(3/2)	,,	0·051	0·0495	F	—	—
(4/2)	,,	0·040	0·0495	F	—	—
(5/2)	,,	0·052	0·0495	F	—	—
(6/2)	,,	0·016	0·096	60	0·577	26·5
(7/2)	,,	0·020	0·096	63	0·613	29·4
(8/2)	,,	0·031	0·0965	57	0·543	41·1
(9/2)	,,	0·040	0·097	F	—	—
(10/2)	,,	0·052	0·052	53	0·499	45·4
(21/2)	,,	0·077	0·095	F	—	—
(22/2)	,,	0·083	0·093	F	—	—
(23/2)	,,	0·105	0·095	F	—	—
(24/2)	,,	0·122	0·096	F	—	—
(11/2)	,,	0·016	0·152	55	0·521	25·25
(12/2)	,,	0·020	0·152	57	0·543	28·7
(13/2)	,,	0·031	0·1515	59	0·566	39·0
(14/2)	,,	0·040	0·152	62	0·601	42·9
(15/2)	,,	0·052	0·151	F	—	—
(25/2)	,,	0·077	0·1505	F	—	—
(26/2)	,,	0·084	0·151	F	—	—
(27/2)	,,	0·102	0·148	F	—	—
(28/2)	,,	0·122	0·150	F	—	—
(16/2)	,,	0·016	0·198	53	0·499	25·8
(17/2)	,,	0·020	0·198	51	0·477	31·0
(18/2)	,,	0·031	0·198	55	0·521	38·3
(19/2)	,,	0·040	0·1985	F	—	—
(20/2)	,,	0·052	0·1985	F	—	—
(29/2)	,,	0·076	0·1985	F	—	—
(30/2)	,,	0·083	0·198	F	—	—
(31/2)	,,	0·100	0·197	F	—	—
(1/3)	0·033	0·016	0·050	70	0·700	40·35
(2/3)	,,	0·020	0·049	67	0·637	50·6
(3/3)	,,	0·031	0·0495	53	0·499	71·2
(4/3)	,,	0·040	0·0495	43	0·394	88·5
(5/3)	,,	0·052	0·0495	47	0·435	95·5
(6/2)	,,	0·016	0·096	75	0·767	38·7
(7/3)	,,	0·020	0·096	71	0·713	36·8
(8/3)	,,	0·031	0·0965	65	0·637	62·0
(9/3)	,,	0·040	0·097	58	0·554	70·0
(10/3)	,,	0·052	0·052	54	0·510	80·9
(21/3)	,,	0·077	0·095	F	—	—
(22/3)	,,	0·083	0·093	F	—	—
(23/3)	,,	0·105	0·095	F	—	—
(24/3)	,,	0·122	0·096	F	—	—
(11/3)	0·033	0·016	0·152	63	0·613	36·5
(12/3)	,,	0·020	0·152	70	0·700	44·8
(13/3)	,,	0·031	0·1515	63	0·601	58·2
(14/3)	,,	0·040	0·152	65	0·637	62·7
(15/3)	,,	0·052	0·151	58	0·554	69·4
(25/3)	,,	0·077	0·1505	52	0·488	82·5
(26/3)	,,	0·084	0·151	F	—	—
(27/3)	,,	0·102	0·148	F	—	—
(28/3)	,,	0·122	0·150	F	—	—
(16/3)	,,	0·016	0·198	72	0·740	37·0
(17/3)	,,	0·020	0·198	65	0·637	46·4
(18/3)	,,	0·031	0·198	68	0·675	56·9
(19/3)	,,	0·040	0·1985	63	0·613	59·9
(20/3)	,,	0·052	0·1985	58	0·554	67·5
(29/3)	,,	0·076	0·1985	54	0·510	74·0
(30/3)	,,	0·083	0·198	48	0·445	75·5
(31/3)	,,	0·100	0·197	48	0·445	83·0
(1/4)	0·042	0·016	0·050	75	0·767	46·4
(2/4)	,,	0·020	0·049	73	0·734	60·0
(3/4)	,,	0·031	0·0495	60	0·577	91·5
(4/4)	,,	0·040	0·0495	67	0·662	113
(5/4)	,,	0·052	0·0405	58	0·554	128
(6/4)	,,	0·016	0·096	83	0·885	44·1
(7/4)	,,	0·020	0·096	79	0·824	55·7
(8/4)	,,	0·031	0·0965	69	0·687	78·1
(9/4)	,,	0·040	0·097	65	0·637	90·4
(10/4)	,,	0·052	0·052	61	0·589	108
(21/4)	,,	0·077	0·095	48	0·445	136
(22/4)	,,	0·083	0·093	45	0·414	142
(23/4)	,,	0·105	0·095	F	—	—
(24/4)	,,	0·122	0·096	F	—	—
(11/4)	,,	0·016	0·152	78	0·810	40
(12/4)	,,	0·020	0·152	77	0·795	53·5
(13/4)	,,	0·031	0·1515	75	0·767	72·5
(14/4)	,,	0·040	0·152	72	0·727	79·6
(15/4)	,,	0·052	0·151	68	0·675	91·2
(25/4)	,,	0·077	0·1505	56	0·532	114
(26/4)	,,	0·084	0·151	48	0·445	126
(27/4)	,,	0·102	0·148	52	0·488	125
(28/4)	,,	0·122	0·150	48	0·455	137
(16/4)	,,	0·016	0·198	83	0·885	41·4
(17/4)	,,	0·020	0·198	80	0·839	55·5
(18/4)	,,	0·031	0·198	75	0·767	71·4
(19/4)	,,	0·040	0·198	72	0·727	76·4
(20/4)	,,	0·052	0·1985	68	0·675	97·5
(29/4)	,,	0·076	0·1985	62	0·601	94·0
(30/4)	,,	0·083	0·198	61	0·589	102
(31/4)	,,	0·100	0·197	53	0·499	114
(1/5)	0·052	0·016	0·050	80	0·839	51·6
(2/5)	,,	0·020	0·040	81	0·854	69·9
(3/5)	,,	0·031	0·0495	67	0·662	115
(4/5)	,,	0·040	0·0495	59	0·566	150
(5/5)	,,	0·052	0·0495	63	0·613	171
(6/5)	,,	0·016	0·096	85	0·916	49·5
(7/5)	,,	0·020	0·096	83	0·884	63·5
(8/5)	,,	0·031	0·0965	75	0·767	95·5
(9/5)	,,	0·040	0·097	75	0·767	114
(10/5)	,,	0·052	0·052	69	0·687	141
(21/5)	,,	0·077	0·095	54	0·510	125
(22/5)	,,	0·083	0·093	50	0·466	192
(23/5)	,,	0·105	0·095	52	0·488	206
(24/5)	,,	0·122	0·096	44	0·404	220
(11/5)	,,	0·016	0·152	86	0·933	44·2
(12/5)	,,	0·020	0·152	86	0·933	60·8
(13/5)	,,	0·031	0·1515	80	0·839	87·5
(14/5)	,,	0·040	0·152	82	0·869	98·5
(15/5)	,,	0·052	0·151	77	0·795	113
(25/5)	,,	0·077	0·1505	58	0·554	147
(26/5)	,,	0·084	0·151	57	0·543	169
(27/5)	,,	0·102	0·148	60	0·577	164
(28/5)	,,	0·122	0·150	55	0·521	178
(16/5)	,,	0·016	0·198	91	1·02	44·4
(17/5)	,,	0·020	0·198	84	0·900	62·4
(18/5)	,,	0·031	0·198	88	0·966	84·4
(19/5)	,,	0·040	0·1985	87	0·869	92·6
(20/5)	,,	0·052	0·1985	74	0·754	111

* F indicates that the bubble failed to open.

TABLE 6—*continued*

Burner No.	Orifice diameter, d, inch	Port diameter, p, inch	Offset radius, r, inch	Cone angle, θ, deg.	Tan ($\theta/2$)	Flow, lb. per hr.
(29/5)	0·052	0·076	0·1985	75	0·767	122
(30/5)	,,	0·083	0·198	80	0·839	134
(31/5)	,,	0·100	0·197	71	0·713	145
(1/6)	0·067	0·016	0·050	89	0·983	55·5
(2/6)	,,	0·020	0·049	83	0·885	78·0
(3/6)	,,	0·031	0·0495	82	0·869	140
(4/6)	,,	0·040	0·0495	70	0·700	205
(5/6)	,,	0·052	0·0495	67	0·662	255
(6/6)	,,	0·016	0·096	93	1·054	52·5
(7/6)	,,	0·020	0·096	94	1·072	72·5
(8/6)	,,	0·031	0·0965	90	1·000	115
(9/6)	,,	0·040	0·097	85	0·916	146
(10/6)	,,	0·052	0·052	75	0·767	186
(21/6)	,,	0·077	0·095	63	0·613	260
(22/6)	,,	0·083	0·093	64	0·625	277
(23/6)	,,	0·105	0·095	53	0·499	300
(24/6)	,,	0·122	0·096	54	0·510	320
(11/6)	,,	0·016	0·152	93	1·054	50·3
(12/6)	,,	0·020	0·152	92	1·036	70·0
(13/6)	,,	0·031	0·1515	87	0·949	110
(14/6)	,,	0·040	0·152	87	0·949	125
(15/6)	,,	0·052	0·151	82	0·869	144
(25/6)	,,	0·077	0·1505	78	0·810	200
(26/6)	,,	0·084	0·151	76	0·781	226
(27/6)	,,	0·102	0·148	74	0·754	220
(28/6)	,,	0·122	0·150	68	0·675	257
(16/6)	,,	0·016	0·198	91	1·018	47
(17/6)	,,	0·020	0·198	69	1·054	69
(18/6)	,,	0·031	0·198	87	0·949	100
(19/6)	,,	0·040	0·1985	88	0·966	115
(20/6)	,,	0·052	0·1985	80	0·839	145
(29/6)	,,	0·076	0·1985	86	0·933	160
(30/6)	,,	0·083	0·198	83	0·855	180
(31/6)	,,	0·100	0·197	73	0·744	206
(1/7)	0·076	0·016	0·050	90	1·000	56·0
(2/7)	,,	0·020	0·049	90	1·000	82·5
(3/7)	,,	0·031	0·0495	86	0·933	160
(4/7)	,,	0·040	0·0495	82	0·869	225
(5/7)	,,	0·052	0·0495	71	0·713	275
(6/7)	,,	0·016	0·096	113	1·511	53·3
(7/7)	,,	0·020	0·096	100	1·192	77·5
(8/7)	,,	0·031	0·0965	94	1·072	130
(9/7)	,,	0·040	0·097	92	1·036	165
(10/7)	,,	0·052	0·052	86	0·933	215
(21/7)	,,	0·077	0·095	72	0·727	290
(22/7)	,,	0·083	0·093	68	0·675	310
(23/7)	,,	0·105	0·095	67	0·662	360
(24/7)	,,	0·122	0·096	62	0·601	370
(11/7)	,,	0·016	0·152	101	1·213	52·2
(12/7)	,,	0·020	0·152	99	1·171	73·2
(13/7)	,,	0·031	0·1515	95	1·091	117
(14/7)	,,	0·040	0·152	96	1·111	138
(15/7)	,,	0·052	0·151	90	1·000	165
(25/7)	,,	0·077	0·1505	80	0·839	217
(26/7)	,,	0·084	0·151	74	0·754	278
(27/7)	,,	0·102	0·148	70	0·700	265
(28/7)	,,	0·122	0·150	77	0·795	280
(16/7)	,,	0·016	0·198	90	1·000	48
(17/7)	,,	0·020	0·198	97	1·130	66·5
(18/7)	,,	0·031	0·198	93	1·054	114
(19/7)	,,	0·040	0·1985	90	1·000	127
(20/7)	,,	0·052	0·1985	88	0·966	166
(29/7)	,,	0·076	0·1985	82	0·869	171
(30/7)	,,	0·083	0·198	86	0·933	195
(31/7)	0·076	0·100	0·197	84	0·900	225
(1/8)	0·100	0·016	0·050	92	1·036	58
(2/8)	,,	0·020	0·049	97	1·130	87·5
(3/8)	,,	0·031	0·0495	88	0·966	183
(4/8)	,,	0·040	0·0495	82	0·869	280
(5/8)	,,	0·052	0·0495	85	0·916	349
(6/8)	,,	0·016	0·096	97	1·130	58
(7/8)	,,	0·020	0·096	114	1·54	83·5
(8/8)	,,	0·031	0·0965	100	1·192	156
(9/8)	,,	0·040	0·097	100	1·192	213
(10/8)	,,	0·052	0·052	91	1·018	270
(21/8)	,,	0·077	0·095	84	0·900	376
(22/8)	,,	0·083	0·093	80	0·839	420
(23/8)	,,	0·105	0·095	74	0·754	461
(24/8)	,,	0·122	0·096	70	0·700	612
(11/8)	,,	0·016	0·152	99	1·171	55
(12/8)	,,	0·020	0·152	93	1·054	79·6
(13/8)	,,	0·031	0·1515	95	1·091	138
(14/8)	,,	0·040	0·152	96	1·111	170
(15/8)	,,	0·052	0·151	97	1·130	205
(25/8)	,,	0·077	0·1505	84	0·900	302
(26/8)	,,	0·084	0·151	83	0·885	362
(27/8)	,,	0·102	0·148	90	1·000	348
(28/8)	,,	0·122	0·150	88	0·966	387
(16/8)	,,	0·016	0·198	112	1·483	50·7
(17/8)	,,	0·020	0·198	114	1·540	80·2
(18/8)	,,	0·031	0·198	100	1·192	135
(19/8)	,,	0·040	0·1985	95	1·092	1·91
(20/8)	,,	0·052	0·1985	94	1·072	207
(29/8)	,,	0·076	0·1985	93	1·054	227
(30/8)	,,	0·083	0·198	90	1·000	263
(31/8)	,,	0·100	0·197	88	0·966	308
(1/9)	0·120	0·016	0·050	99	1·171	59·5
(2/9)	,,	0·020	0·049	88	0·966	100
(3/9)	,,	0·031	0·0495	91	1·018	200
(4/9)	,,	0·040	0·0495	92	1·036	300
(5/9)	,,	0·052	0·0495	87	0·949	362
(6/9)	,,	0·016	0·096	144	1·540	58
(7/9)	,,	0·020	0·096	98	1·150	88
(8/9)	,,	0·031	0·0965	99	1·171	173
(9/9)	,,	0·040	0·097	96	1·111	230
(10/9)	,,	0·052	0·052	99	1·171	303
(21/9)	,,	0·077	0·095	86	0·935	454
(22/9)	,,	0·083	0·093	86	0·933	495
(23/9)	,,	0·105	0·095	74	0·754	555
(24/9)	,,	0·122	0·096	73	0·740	601
(11/9)	,,	0·016	0·152	104	1·280	56
(12/9)	,,	0·020	0·152	104	1·280	81
(13/9)	,,	0·031	0·1515	104	1·280	160
(14/9)	,,	0·040	0·152	102	1·235	190
(15/9)	,,	0·052	0·151	104	1·280	238
(25/9)	,,	0·077	0·1505	92	1·036	368
(26/9)	,,	0·084	0·151	86	0·933	438
(27/9)	,,	0·102	0·148	88	0·966	426
(28/9)	,,	0·122	0·150	81	0·854	569
(16/9)	,,	0·016	0·198	100	1·192	57
(17/9)	,,	0·020	0·198	106	1·327	83
(18/9)	,,	0·031	0·198	104	1·280	147
(19/9)	,,	0·040	0·1985	98	1·150	175
(20/9)	,,	0·052	0·1985	98	1·150	240
(29/9)	,,	0·076	0·1985	92	1·036	261
(30/9)	,,	0·083	0·198	92	1·036	321
(31/9)	,,	0·100	0·197	92	1·036	370

spill line the amount of fuel spilt can be altered. This affects the output of the atomizer in two ways. As the spill line is opened and the spill pressure reduced, less fuel flows through the orifice, d, and the cone angle of the spray also increases, even though the fuel supply pressure is held constant. The fuel delivery can thus be varied by a ratio which is a function of the ratio of spill diameter, S, to final orifice diameter, d. Fig. 10 shows the relation of these ratios according to Simons. A combination of fuel control by (1) varying fuel supply pressure, and (2) varying spill pressure, permits a fuel-flow variation of 100 to 1, or more.

ATOMIZER SHAPE AND CONE ANGLE

Figs. 11–13 show three of the very many curves which may be drawn to represent the measurements of cone angle quoted in Table 4. They show the relation between tan ($\theta/2$) and (1) the orifice diameter, (2) the port diameter, and (3) orifice and port diameter varied together. It is interesting to note that the cone angle increases with orifice diameter and decreases as port diameter increases. Increasing both diameters together causes an increase in the cone angle. Plots of the cone-angle measurements reveal certain minor inconsistencies which are due to the

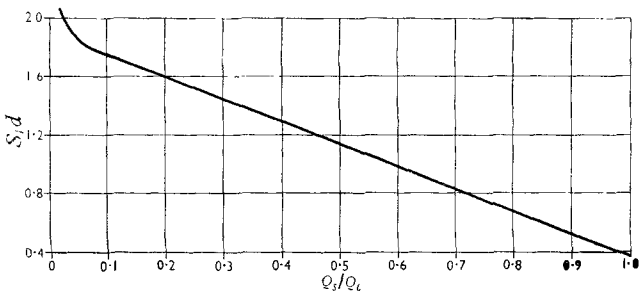


Fig. 10. Effect of Spill Diameter on Flow

Q_s Flow with spill fully open.
Q_c Flow with spill completely closed.

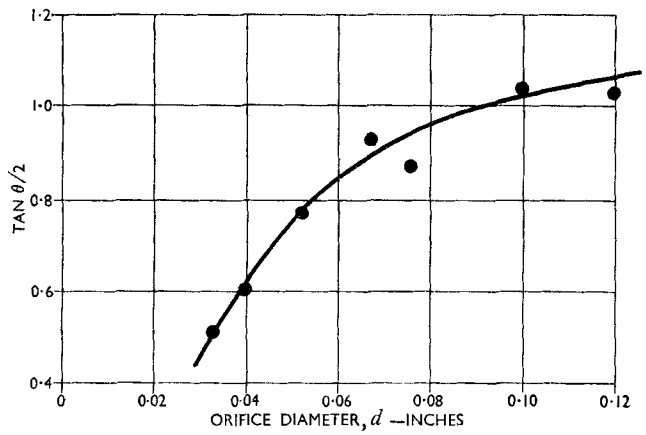


Fig. 11. Variation of Cone Angle, θ, with Diameter of Final Orifice, d

For atomizers of 0·5 inch chamber diameter, 0·077 inch port diameter, and 0·198 inch offset radius.

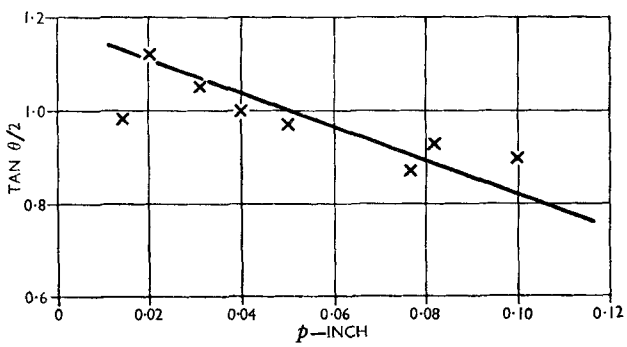


Fig. 12. Variation of Cone Angle, θ, with Port Diameter, p

For atomizers of 0·5 inch chamber diameter, 0·198 inch offset radius, and diameter of final orifice of 0·077 inch.

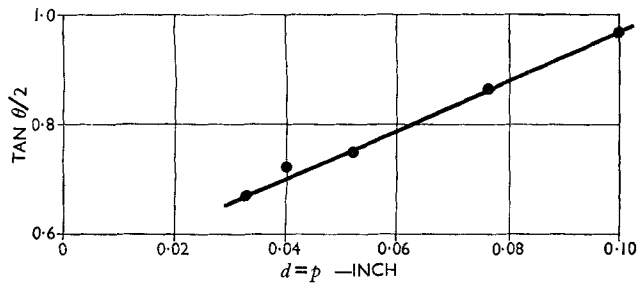


Fig. 13. Variation of Cone Angle, θ, with Port Diameter, p, and Diameter of Final Orifice, d, Changed Together

difficulty of ensuring the desired finish on the various parts of the atomizer. In particular, it is difficult to achieve satisfactory holes of a few thousandths of an inch in diameter.

SPRAY PARTICLE SIZE

The size of the droplets produced by a swirl atomizer generally has a distribution which conforms with the equation

$$R = e^{-(x/\bar{x})^n} \quad . \quad . \quad . \quad . \quad . \quad . \quad (3)$$

where R is the proportion by weight of the spray of particles of diameter greater than x; $\bar{x}$ and n are constants determining, respectively, the general size of the droplets and the distribution pattern. Needham (1946) examined the relation between surface mean diameter and $\bar{x}$ for Lucas atomizers of different size and shape and was able to show that there was good correlation between particle size and 182 $Q^{0\cdot25}/P^{0\cdot4}$. The test results were in the pressure range 6–125 lb. per sq. in., and the flow numbers varied between 0·5 and 4·15. Needham also examined the relation between particle size, flow, and pressure for a Power Jets atomizer of final orifice d = 0·040 inch, tangential port diameter p = 0·028 inch, spill orifice 0·060 inch, and chamber diameter 0·2 inch. Making no distinction between conditions with the spill line open or closed he deduced that the diameter of the droplet which has the same surface to volume diameter as the spray, sometimes called the surface mean diameter (S.M.D.), was

$$\text{S.M.D.} = 360\frac{Q^{0\cdot3}}{P^{0\cdot5}} \quad . \quad . \quad . \quad . \quad . \quad (4)$$

When re-examined the results revealed that the relation

$$\text{S.M.D.} = 325\frac{Q^{0\cdot318}}{P^{0\cdot530}} \quad . \quad . \quad . \quad . \quad . \quad (5)$$

more accurately represented the relation between S.M.D., Q, and P for the Power Jets atomizer with spill closed (Fig. 14). Fig. 15 shows the effect, for the same atomizer A, of opening the spill line on the relation between (1) the ratio of the S.M.D. of the spray produced by the atomizer with spill open to the S.M.D.

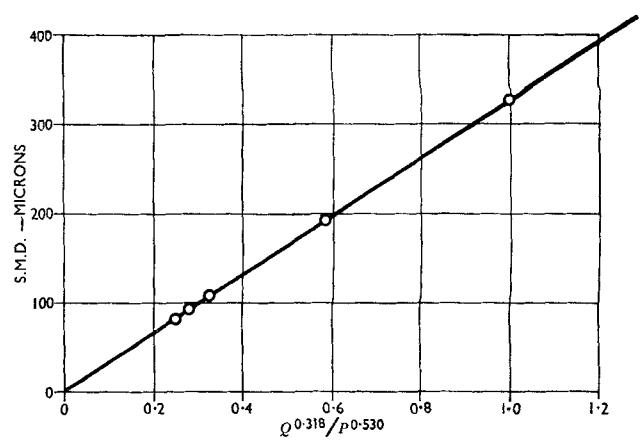


Fig. 14. Variation of Surface Mean Diameter (S.M.D.) with Fuel Flow Q in lb. per hr. and Pressure P in lb. per sq. in.

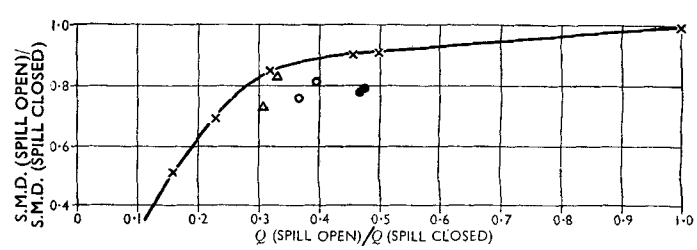


Fig. 15. Variation of Surface Mean Diameter (S.M.D.) with Fuel Output when Flow is Altered by Using Spill Control

× Atomizer A. △ Atomizer C.
○ Atomizer B. ● Atomizer D.

of the spray produced when the spill was closed, and (2) the corresponding flow ratios. Since Needham has shown that equation (4) held for Lucas atomizers of different shape and flow number, it is reasonable to expect that equation (5) will hold for Power Jets atomizers of different flow number and different shape. There are no other data on particle size available concerning the performance of Power Jets atomizers with spill closed. There are, however, six measurements made by Joyce which refer to three atomizers of almost standard shape and of orifice diameter 0·036, 0·050, and 0·075 inch, which are here referred to as atomizers B, C, and D, respectively. Using equation (5) to calculate the S.M.D. with spill closed, it is possible to determine the ratio of S.M.D. with spill open to S.M.D. with spill closed, and plot these values against the corresponding ratios of fuel flow with spill line open to fuel flow with spill closed. This has been done in Fig. 15. The fact that the points are so close to the curve through the crosses indicates that some reliance may be placed on equation (5) and the curve of Fig. 15 to calculate the S.M.D. for spill atomizers. Even so, it is obviously desirable to do tests on more atomizers. According to curves published by Joyce (1949), it may be deduced that the S.M.D. of a swirl spray increases only slightly with viscosity—probably as the kinematic viscosity raised to the power 0·2.

CONCLUSION

It is thus shown that the Power Jets type of atomizer can be designed to give the appropriate output for any fuel of low enough viscosity (*circa* 20 centipoises) and that the effect of varying the atomizer shape can be predicted. Similarly the atomizer spray particle size may be estimated when it is known that the air into which the fuel is sprayed has a pressure of 1 atm.

Further work on the effect of the combustion air pressure on atomization and work on the atomization by spill atomizers, especially for fuels of relatively high viscosity, will also be valuable.

ACKNOWLEDGEMENT

This paper describes part of a programme of work at the National Gas Turbine Establishment and acknowledgement is made to the Chief Scientist, Ministry of Supply, for permission to publish. Acknowledgement is also made of the work of Mr. H. H. Heath who made the measurements of cone angle of the 279 atomizers of different shape.

APPENDIX

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Communications

Mr. D. R. Carlisle, B.Sc. (Derby), wrote that the first section of the author's paper was valuable in confirming that, with atomizers of similar geometrical design, using a range of different fluids, a unique relation between discharge coefficient $\sqrt{(Q^2/d^4P\rho)}$ and $\mathcal{R}_e$. $(Q/\mu d)$ existed.

He had not, however, shown how to design an atomizer to give a required flow and spray angle, although, apparently, that was why the work described in the second section had been carried out.

The 279 atomizer combinations covered such a range of geometrical proportions, many of them being wide of practical requirements, that useful empirical relations might have been masked.

A criticism of the atomizers was that only seventy-two had swirl ports approximating to tangential and of those about half had D/d ratios greater than 10. Indications were that D should not be greater than $5d$ for an efficient atomizer. In addition, orifice length had been found to affect spray angle and losses in pressure energy critically, and for that reason should be kept as short as possible, for example, $d/5$.

Described below was work carried out on behalf of his own company on an attempted flow and spray-angle correlation of swirl atomizers and the results from that work were compared with the author's atomizer calibrations.

An Attempted Correlation of Swirl Atomizers Using Kerosene of Specific Gravity 0·8. Taylor's (1948)* inviscid theory for atomizers showed that the discharge coefficient, K, was a unique

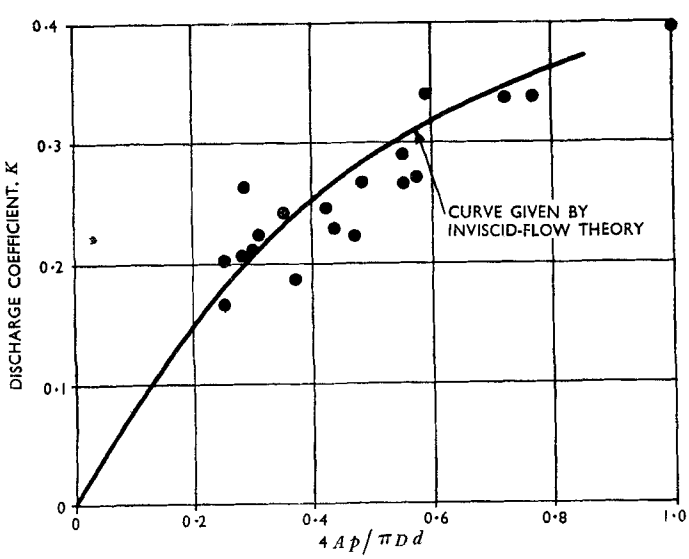

Fig. 16. Discharge Coefficient of Nineteen Atomizers at 700 lb. per sq. in.

function of the parameter, area of inlet ports divided by the product of swirl chamber radius and orifice radius ($4Ap/\pi Dd$ in Figs. 16–26) and also that spray angle was a unique function of K. In practice it was found that the use of a viscous fluid modified the flow patterns within the swirl chamber and caused substantial deviations from the ideal case, as, in fact, the author's work showed.

Twenty atomizers in current use had been selected for tests using kerosene; they ranged in flow number from 0·1 to 9·25 and in spray angle from 70 to 107 deg. They varied in geometrical form, some having drilled ports and others rectangular slots.

Those had each been tested for flow and spray angle at pressures ranging from 5 to 700 lb. per sq. in.

Flow Parameter. Fig. 16 showed the plot of discharge coefficient, K, measured at 700 lb. per sq. in. The author had

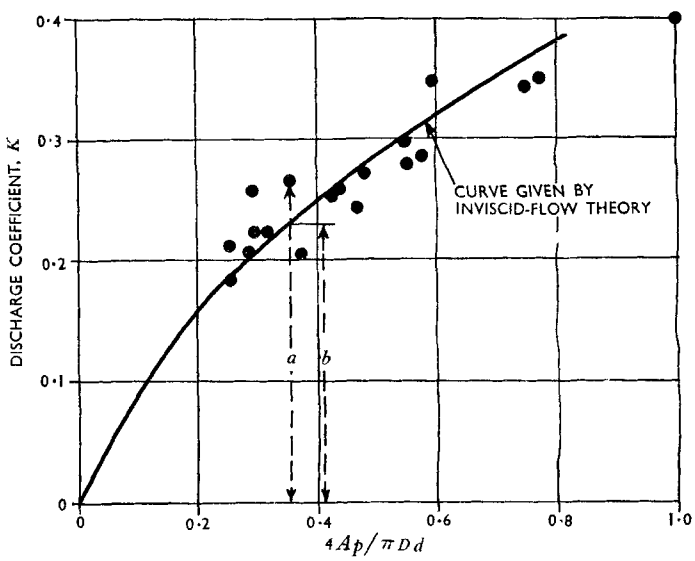

Fig. 17. Discharge Coefficient at Same $\mathcal{R}_e$.

shown how K varied with $\mathcal{R}_e$ and so, in Fig. 17, K had been plotted for each atomizer at a pressure giving the same $\mathcal{R}_e$. Those results showed slightly less scatter.

An examination of the scatter showed that those with a high D/d ratio gave higher values of K than those with a low D/d ratio.

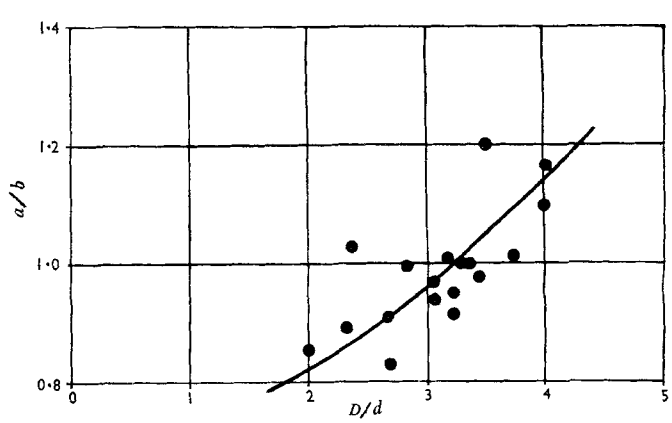

Fig. 18. Deviation of K from Ideal Curve Plotted Against Ratio of Swirl Chamber Diameter to Orifice Diameter, D/d

a/b is the ratio of measured K to K from ideal curve.

Thus, in Fig. 18, the proportional deviation of K from the ideal curve had been plotted against the ratio D/d and exhibited a definite tendency. A mean line drawn through those points had been used to correct K and the new values plotted as K' in

* Taylor, G. I. 1948 Proc. Seventh International Congress of Applied Mechanics, vol. 2 (1), p. 280, 'The Mechanism of Swirl Atomisers'.

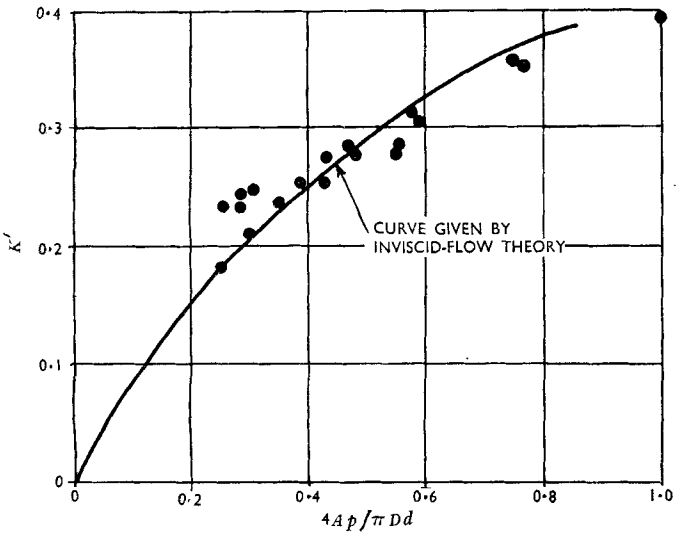

Fig. 19. K Corrected for D/d Ratio (K') Plotted Against $4Ap/\pi Dd$

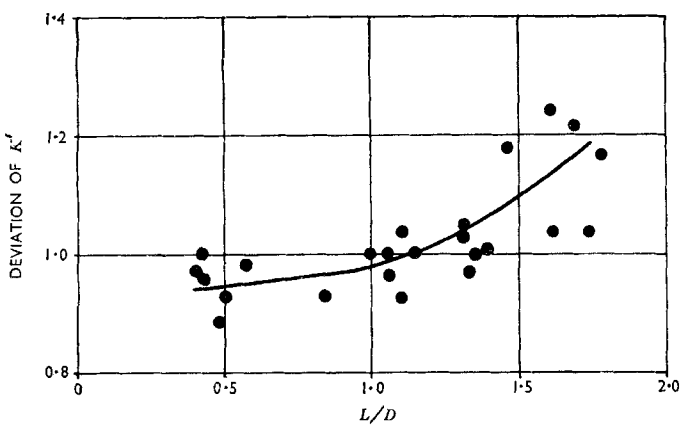

Fig. 20. Proportional Deviation of K' from Ideal Curve Plotted Against Ratio L/D

L Total length of swirl chamber.

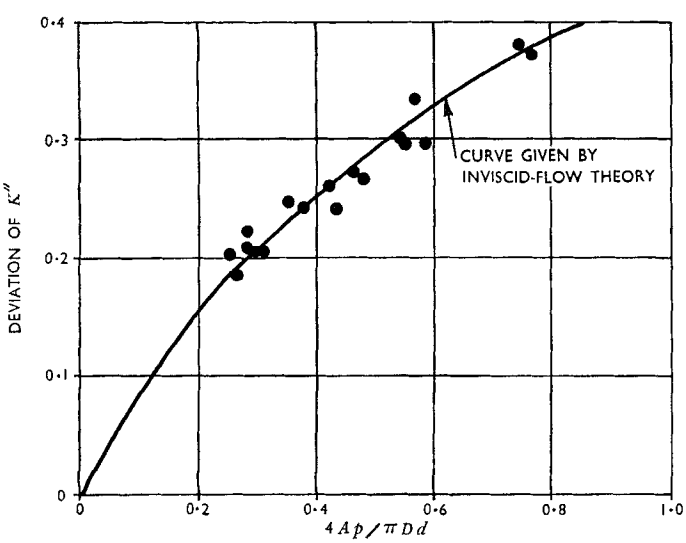

Fig. 21. Final Corrected K (K'') Plotted Against $4Ap/\pi Dd$

K'' is K' corrected for L/D ratio.

Fig. 19. (It was to be noted that the D/d ratio which lay on the ideal line was 3·5.) The points had less scatter than previously.

A separate experiment had shown that increasing the swirl-chamber length increased the discharge coefficient, and so in Fig. 20 the deviation of K' from the ideal curve had been plotted against the ratio of total swirl-chamber length to swirl-chamber diameter, L/D. That also showed a definite tendency and again a mean curve was used to correct K' for L/D ratio effect.

Fig. 21 showed the final corrected discharge coefficients, K'', and the scatter was now within reasonable limits.

The corrections for D/d ratio and L/D ratio could conveniently be combined together on one curve.

Spray Angle. Spray angle was a difficult quantity to measure. At lower pressures after the tulip had disappeared, the spray angle gradually increased with pressure to a maximum and then decreased owing to air entrainment. The pressure at which the maximum value occurred varied from nozzle to nozzle. The procedure in those tests was to take the maximum value irrespective of pressure.

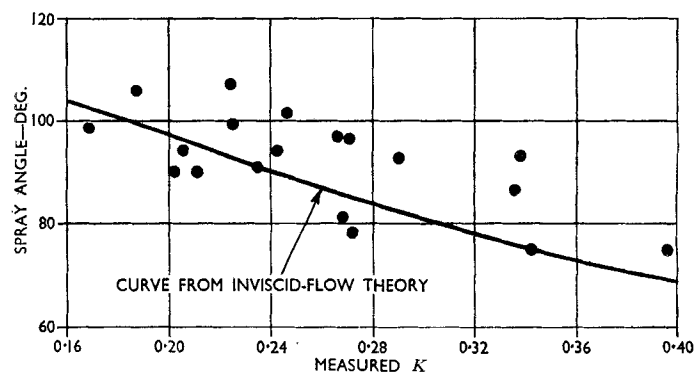

Fig. 22. Plot of Measured Spray Angle Against Measured Discharge Coefficient

The spray angles were shown in Fig. 22 plotted against measured discharge coefficient. Examination showed a tendency for small orifice diameters to have lower spray angles than large orifice diameters. The best correlation, however, was found by plotting deviation from the ideal curve against orifice diameter multiplied by discharge coefficient ($K \times d$). That latter function was a rough measure of film thickness within the orifice and

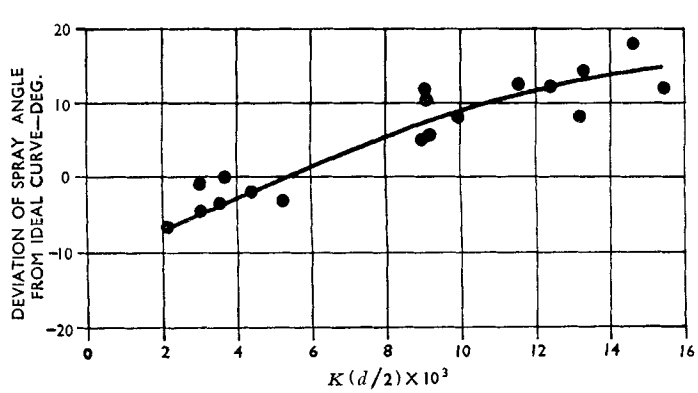

Fig. 23. Deviation of Spray Angle from Ideal Curve Plotted Against $K \cdot d/2$

would be proportional to $\mathcal{R}e$ based on film thickness if the pressure was constant. That was shown in Fig. 23 and the corrected spray angles in Fig. 24.

To Design an Atomizer of Given Spray Angle and Flow Number Using Kerosene. From Fig. 24 the value of K corresponding to

the required spray angle should be read off and orifice diameter calculated from $d = 0{\cdot}0245\sqrt{(FN/K)}$ inches.

Then the spray angle deviation should be obtained from Fig. 23 and a new value of K obtained from Fig. 24; d which ought then to be sufficiently near the required value (or continue successive approximation) should be recalculated.

A swirl-chamber diameter not greater than $5d$ should, if possible, be selected and with the shortest convenient swirl-chamber length the correction factors for K should be obtained from Figs. 18 and 20. With the corrected value of K, $4Ap/\pi Dd$ should be read off from Fig. 21. That gave the swirl port area. Thus, the various atomizer dimensions were then established

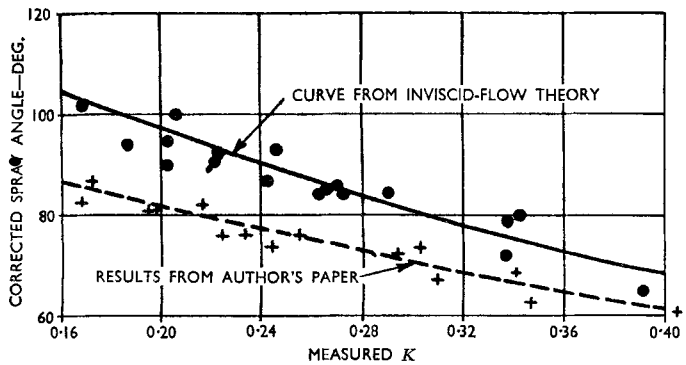

Fig. 24. Spray Angles Corrected to Ideal Curve by $K \cdot d/2$ Factor Plotted Against Measured Discharge Coefficient

to give a required flow number and spray angle, the flow number being at a pressure equal to $(0{\cdot}5/d)^2$ (that being the $\mathcal{R}_e$ selected to plot Fig. 17).

Examination of Author's Results in Table 6. Of the atomizer combinations of Table 6 only those with an offset radius of 0·2 inch and those with a developed spray at 100 lb. per sq. in. had been selected. Most of those with a value of $4Ap/\pi Dd$ below 0·1 had been excluded. In addition, those with sleeve No. 29, which was not consistent with the others, had been excluded.

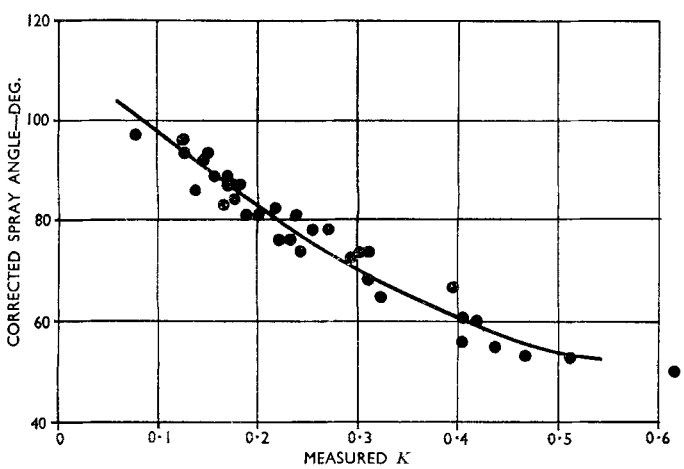

Fig. 25. Plot of Corrected Spray Angles Against Measured Discharge Coefficient for Thirty-six Atomizers Selected from Author's Paper

If those results (thirty-six atomizers) were corrected for spray angle as in Fig. 23 a very good correlation between corrected spray angle and measured discharge coefficient was obtained as shown in Fig. 25. The curve was also shown in Fig. 24 in order to compare it with the results of his own company and it could be seen that the author's spray angles lay some 7–17 deg. below. The difference could be accounted for by:

(1) Difference in orifice length;
(2) Method of measurement.

Whilst that spray-angle correlation was very satisfactory, the flow parameters did not give such a clear-cut relation. The results were shown in Fig. 26. It could be seen that each orifice size (or D/d value) had its own curve of K against $4Ap/\pi Dd$. That substantiated qualitatively the correction for D/d in Fig. 18. It had also to be remembered that all those points were at 100 lb. per sq. in. pressure and could not, therefore, be plotted at the same $\mathcal{R}_e$. However, that family of curves could be used as they stood but tests using more swirl-chamber diameters and

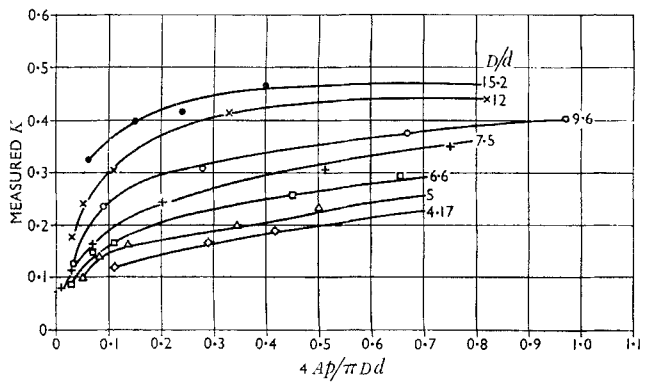

Fig. 26. Flow Parameters: Author's Results for Measured K Plotted Against $4Ap/\pi Dd$

more values of $\mathcal{R}_e$ would be valuable to substantiate them. Possibly, the curves might be correlated to one unique curve if $\mathcal{R}_e$ based on fluid film thickness within the orifice were used.

Summary. It had been shown that swirl atomizers for most practical purposes could be designed to give a required flow number and spray angle, with reasonable accuracy, using the inviscid theory and corrections for D/d ratio, L/D ratio, and $\mathcal{R}_e$.

The correlation between spray angle, corrected for $Kd/2$, and K was particularly valuable and the author's results and those he himself had described could most probably be reconciled.

The author's results of K against $4Ap/\pi Dd$ could be used for design with less limitations than the curves given in Figs. 16–26 if they were augmented by more tests using different swirl-chamber diameters. A wider pressure range should also be used.

Further Observations. It was true as the author had pointed out that flow number varied with pressure but at higher values of $\mathcal{R}_e$ flow number remained substantially constant and was, therefore, very useful.

It should be emphasized that the correcting factor $Kd/2$ for correlating spray angle with K only applied to kerosene, but it might well be that that factor, if put in terms of true $\mathcal{R}_e$ based on orifice film thickness, could be used for other fluids. Further work on that would be useful.

Work on swirl atomizers could profitably be directed towards achieving understanding of the losses within the swirl chamber which could amount to more than 50 per cent of the total pressure at inlet.

Mr. R. H. B. Forster, B.Sc. (Eng.) (*Member*), wrote that in many types of swirl atomizer some care was taken to make the outer edge of the inlet port truly tangential with the swirl chamber so that as little disturbance as possible was caused at the inlet. In fact, the details of the atomizer of standard shape which the author had quoted appeared to be determined with that object in view.

It was therefore surprising that the design selected by the author for experiment departed so radically from that principle.

There were only nine out of the 279 tests for which that condition held. He would be interested to know whether the author considered that that had any effect. The conclusion to be drawn from the result with small nozzles ran counter to that.

If the offset radius was increased the tangential velocity, U, at the orifice increased in accordance with the relation Ur^2 was a constant. The effect of that in widening the cone angle and decreasing the flow slightly was seen clearly in Table 6. Without losses due to shock and friction, increase of offset radius should improve atomization. With the smaller nozzles tested, increase of the offset angle up to a certain point had the effect of causing the bubble to open, but with further increase of offset the bubble failed to open. That was obviously due to the greater loss of energy owing to the much longer path the liquid had to travel, in those circumstances, with small ports and high speed.

It was somewhat unfortunate that the author had found it necessary to employ units which were so inconsistent. He had stated clearly that they were the units likely to be used in Britain by users of fuel atomizers. However, that type of nozzle was used for a variety of other liquids where different units might be employed and, in addition, its use outside Britain was rendered difficult.

In those circumstances, there was much to be said for the employment of a dimensionless parameter.

In the author's units the quantity Q^2/d^4P had the dimensions (lb. or gramme) × (second^2 or hour^2) × (cm.^{3} or in.^{2} × feet).

Those were not simple to convert into a consistent set.

Mr. R. P. Fraser, O.B.E., Dr. P. Eisenklam, Dr. N. Dombrowski, and Mr. D. Hasson (London) wrote that the paper provided some very useful information on the effect of flow properties of two different designs of pressure atomizer. First, the author had considered the effect of liquid properties on a number of scale models of a 'standard' atomizer (Table 1), and secondly, the effect of varying the dimensions of atomizers of 'non-standard' type (Tables 4 and 5). His use of the word 'standard' was, perhaps, unfortunate, since the design of swirl atomizer employed was not commonly used by other investigators, or in industry. No standards yet existed.

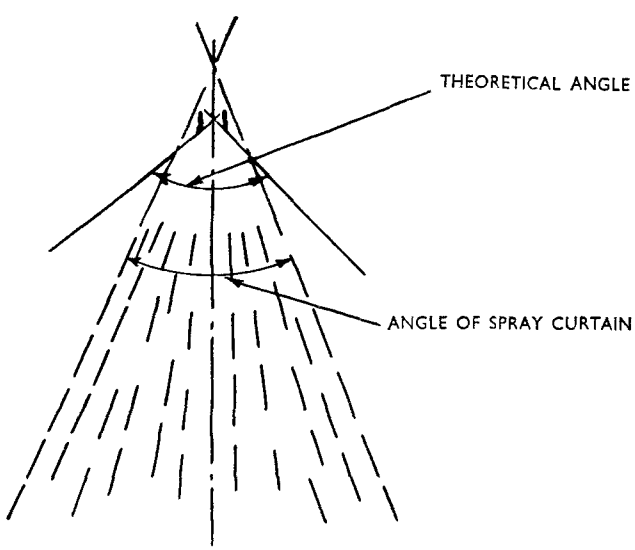

Fig. 27. Theoretical Spray Angle

The correlation of liquid properties with discharge had not been related to atomizers of 'non-standard' dimensions, and equation (2), which gave the correlation with design parameters, was of limited use since their interactions were not considered.

In the 'standard' atomizer the spray angles were not given for the various flows through the atomizer. It was, therefore, difficult to ascertain the general effect of both liquid properties and atomizer dimensions. Theories of flow defined the theoretical spray angle as that at which the tangent to the curving conical liquid sheet attained its maximum value (Fig. 27). The author, and many other investigators, measured an apparent angle which was influenced by the properties of the spray curtain, such as the diffuse boundary or the spatial distribution of the drops, instead of the theoretical angle. Such measurements were dependent on the ambient atmosphere and could be compared only under similar conditions. They also gave rise to anomalous results when compared with ideal theory. Thus, according to ideal theory, the spray angle decreased with increasing $p^2/2rd$, and Figs. 11 and 12 were in agreement with that, but Fig. 13 was at variance. That discrepancy could be explained by the reduced effect of air resistance with the nozzles of higher flow number, where the reduction in the sheet cone angle was more than counterbalanced by the increased momentum of the drops so that the apparent flow angle would increase.

The author had stated that equation (5) for the drop size, which was based on Needham's results (1946), should hold for atomizers of different flow number and different shape, and should therefore be applicable to the nozzles examined. Needham, however, had analysed sprays of approximately constant cone angle. Since the cone angle affected the drop size, and the author's results showed a considerable variation in cone angle, the application of equation (5) could be an approximation only, and particularly so at low pressures.

According to ideal theory, the discharge coefficient $K_Q = \dfrac{Q}{\rho\frac{\pi}{4}d^2\sqrt{\dfrac{2gP}{\rho}}}$, was primarily a function of the parameter $p^2/2rd$. For the 'standard' atomizers the parameter $p^2/2rd$ had a constant value of 0·167, which corresponded to a theoretical discharge coefficient of 0·13. It was known that frictional effects affected the value of the discharge coefficient. From the author's correlation $\dfrac{Q^2}{d^4P\rho} = f\left(\dfrac{Q}{\mu d}\right)$, it followed that $K_Q{}^2 = \Phi(K_Q \times \mathcal{R}e)$, where $\mathcal{R}e$ was the Reynolds number based on the orifice diameter. Thus Fig. 3 could be replotted to show the discharge coefficient K_Q as a function of $\mathcal{R}e$ (Fig. 28). At the lowest flow, $K_Q = 0{\cdot}66$,

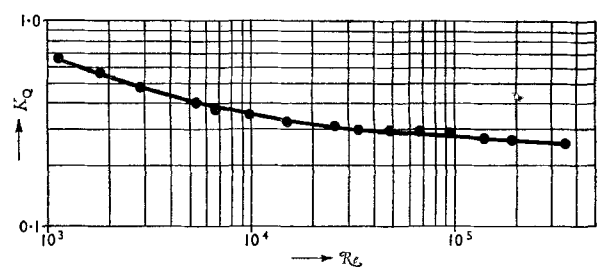

Fig. 28. Replotted Fig. 3 Showing Discharge Coefficient, K_Q, as Function of $\mathcal{R}e$

and in the range where the atomizer operated satisfactorily, K_Q varied between 0·4 and 0·25. It was seen that the theoretical value was never reached.

The remarks made on the basis of Fig. 4, that the flow number changed with pressure, density, and viscosity, were also a result of frictional effects. The flow number, which was given by the expression $FN = \dfrac{Q}{\rho\sqrt{P}} = K_Q\frac{\pi}{4}d^2\sqrt{\dfrac{2g}{\rho}}$, depended on K_Q, and therefore varied with the Reynolds number as shown for the examined atomizers in Fig. 28.

It was hoped, in due course, to publish the results of an investigation of the effect of friction on the flow characteristics in swirl atomizers of different designs which was proceeding in the High Speed Fluid Kinetics Laboratory of the Department of Chemical Engineering and Applied Chemistry of the Imperial College of Science and Technology.

Mr. B. E. Knight, M.A. (London), wrote that data additional to those available to Needham (1946)—quoted in the paper—had been analysed by him to give an empirical formula for spray drop size (S.M.D.) that included the effect of fuel viscosity. An

attempt had also been made to obtain the effect of surface tension and gas viscosity by putting the variables in non-dimensional form.

The formulae obtained were

$$d = 220P^{-0\cdot458}Q^{0\cdot209}\nu_L^{0\cdot215} \quad . \quad . \quad . \quad . \quad (6)$$

$$S \propto R_L^{-0\cdot215}W^{-0\cdot115}R_G^{-0\cdot251} \quad . \quad . \quad . \quad . \quad (7)$$

where d = Sauter mean diameter, in microns; P = pressure drop, in lb. per sq. in.; Q = mass flow, in lb. per hr.; ν_L = kinematic viscosity of fuel, in centistokes; ν_G = kinematic viscosity of gas, in centistokes; ρ_G = gas density, in grammes per cu. cm.; γ_L = surface tension, in dynes per cm.; F = flow number, in gal. per hr. per lb. per sq. in.; $S(\propto d/F^{\frac{1}{2}})$ = non-dimensional form for drop size; $R_L(\propto P^{\frac{1}{2}}F^{\frac{1}{2}}/\nu_L)$ = Reynolds number for fuel; $R_G(\propto P^{\frac{1}{2}}F^{\frac{1}{2}}/\nu_G)$ = Reynolds number for the gas; and $W(\propto \rho_G PF^{\frac{1}{2}}/\gamma_L)$ = Webers number based on the gas density.

Those results had been obtained by weighting the smoothed data of Lubbock and Bowen (1948)* equally with the other measurements on Lucas burners. Table 7 showed the agreement between Needham's equation (columns A) and equation (6) (columns B), with ν_L = 2·3 centistokes.

TABLE 7. S.M.D., MICRONS

P, lb. per sq. in.	Q, lb. per hr.			
	2		200	
	A	B	A	B
50	45	52	143	135
1,000	14	13	43	35

In the non-dimensional analysis the well-known forms for S, R_L, R_G, and W could be obtained by substituting a diameter of the atomizer for $F^{\frac{1}{2}}$ and a velocity for $P^{\frac{1}{2}}$. The actual choice of variables had practical and theoretical convenience that more than offset the need for a fractional power of the fuel density in the factor of proportionality. That had been omitted to facilitate comparison with the standard forms and because a full discussion could not be given.

The attempt to estimate the effect of surface tension and gas viscosity had been partially frustrated because of a fairly close correlation between W and R_G arising from the small range of flow numbers in the data. The corresponding exponents in equation (7) were not significant individually and could be varied over a wide range, on a one-greater one-smaller basis, without spoiling the fit to the data. Reducing the exponent of W to zero led to an exponent of R_G so large as to be improbable. Reducing the exponent of R_G to zero led to the expression

$$S \propto R_L^{-0\cdot211}W^{-0\cdot257} \quad . \quad . \quad . \quad . \quad . \quad (8)$$

Subsequent results, communicated privately†, showed an effect of surface tension between the limits given by the expressions (7) and (8).

The following example showed that equation (6) gave a good idea of the atomization of plain holes such as were used in Diesel engines.

P = 3,520 lb. per sq. in.
ν_L = 5 centistokes.
Hole diameter = 0·38 mm.
Calculated S.M.D. = 21·2 microns
Wax spray measurements

hole length 0·4 mm. = 21·3 microns.
hole length 0·9 mm. = 17·9 microns.

* LUBBOCK, I., and BOWEN, I. G. 1948 Proc. Inst. of Fuel and Inst. of Petroleum Joint Conference on Modern Applications of Liquid Fuels, p. 231.

† Private communication from the Shell Petroleum Co., Ltd., Thornton Research Centre.

Professor H. B. SQUIRE (London) wrote that an advance in understanding of the operation of swirl atomizers could be obtained by measurements of the fluid velocity or film thickness. A scaled-up model might be used for such measurements.

He urged that consistent units be used for future work in the field.

Mr. R. WOOD, B.Sc. (*Graduate*), wrote that the paper gave a great deal of information on the performance of atomizers of different shapes and sizes; in particular, Figs. 3 and 4 and equation (5) were most useful for predicting the behaviour of swirl atomizers. It was unfortunate, however, that many of the atomizers whose performances were listed in Table 6 had shapes which were not normally used in practice. In some instances the offset radius of the tangential ports was less than the radius of the discharge orifice, and at the other extreme the offset radius was frequently more than ten times the orifice radius. Even the atomizers of standard shape had a relatively large swirl-chamber diameter, giving an unnecessarily large wetted surface in the vortex chamber. Table 3, which showed the main dimensions of those atomizers, gave impossible values, in the final column, for the offset radii of the swirl ports.

In regard to equation (2), he would be interested to have the author's opinion on what the quantity 'l' represented in an atomizer. Its significance was rather obscure, and it seemed that that quantity had been introduced merely to reconcile equation (2) with Fig. 3. One linear dimension of the atomizer which, surprisingly, had been ignored in the analysis of shape effect was the offset radius of the tangential ports. An examination of the figures in Table 6 revealed that, for an atomizer of standard shape, $Q \propto r^{-0\cdot84}$. The effect of shape on output was then given by the expression

$$Q = Ad^{0\cdot94}p^{0\cdot70}r^{-0\cdot84} \quad . \quad . \quad . \quad . \quad . \quad (9)$$

and there was evidently no connexion therefore between l and r.

Apart from the orifice length, the three dimensions d, p, and r described the shape of the atomizer completely, provided that, as in that instance, the angle of the vortex chamber was constant, and the significance of another linear dimension (l) was not clear. He would value the author's comment on that point.

Equation (2), or alternatively equation (9), is strictly applicable only to atomizers having the author's standard shape, and would not always give an even approximate idea of the effect of changing shape if the shape was very different. For an atomizer of more usual shape (d = 0·42 inch, p = 0·52 inch, r = 0·96 inch) the expression of shape effect became

$$Q = A_1d^{1\cdot21}p^{0\cdot61}r^{-0\cdot24} \quad . \quad . \quad . \quad (10)$$

In that event the effect of tangential port pitch was much reduced.

AUTHOR'S REPLY

Mr. A. RADCLIFFE wrote in reply that he thanked those who had contributed communications. He did not think that the data which enabled the effect of chamber shape to be determined was buried quite so deeply as Mr. D. R. Carlisle had suggested. He thought that Mr. Carlisle's method of using the data was a good one.

He wished to assure Mr. Forster that he realized that many of the 279 atomizer shapes were unrealistic. They had been used to show what happened when the design was taken to extremes. He agreed with Mr. Forster and Professor Squire that the use of consistent units was desirable and had been pleased to be able to use them in Fig. 3. It was unfortunate that in other places the general use of consistent units would have made the paper difficult to read.

The author was grateful to Mr. Frazer, Dr. Eisenklam, Dr. Dombrowski, and Mr. Hasson for their elaboration of the idea that atomizer flow was to some extent controlled by viscosity or friction. No doubt they and Mr. Carlisle would agree that it was surprising that the inviscid theory (Taylor 1948) gave such a good picture of the dependence of discharge coefficient

and cone angle upon p^2/dr even though the flow inside the atomizer was well described by a viscous theory of flow (Taylor 1950).

The author considered that cone angle—except when near to zero or 180 deg.—was not important as a parameter controlling particle size. It was pleasant to be able to neglect something. As Mr. Knight had shown there were many factors which might affect particle size.

Mr. Wood had asked about the parameter l. That had the dimension of length and took into account all the other dimensions which did not vary with d and p. It was assumed to include surface finish.